

Site Validation Report

Sunflower Solar

391 Brockington Rd.
Kingstree, SC 29556

SITE

AC System Size	DC System Size	Valid Data Date
10000	13882	12/19/2023

PRODUCTION HEIRARCHY

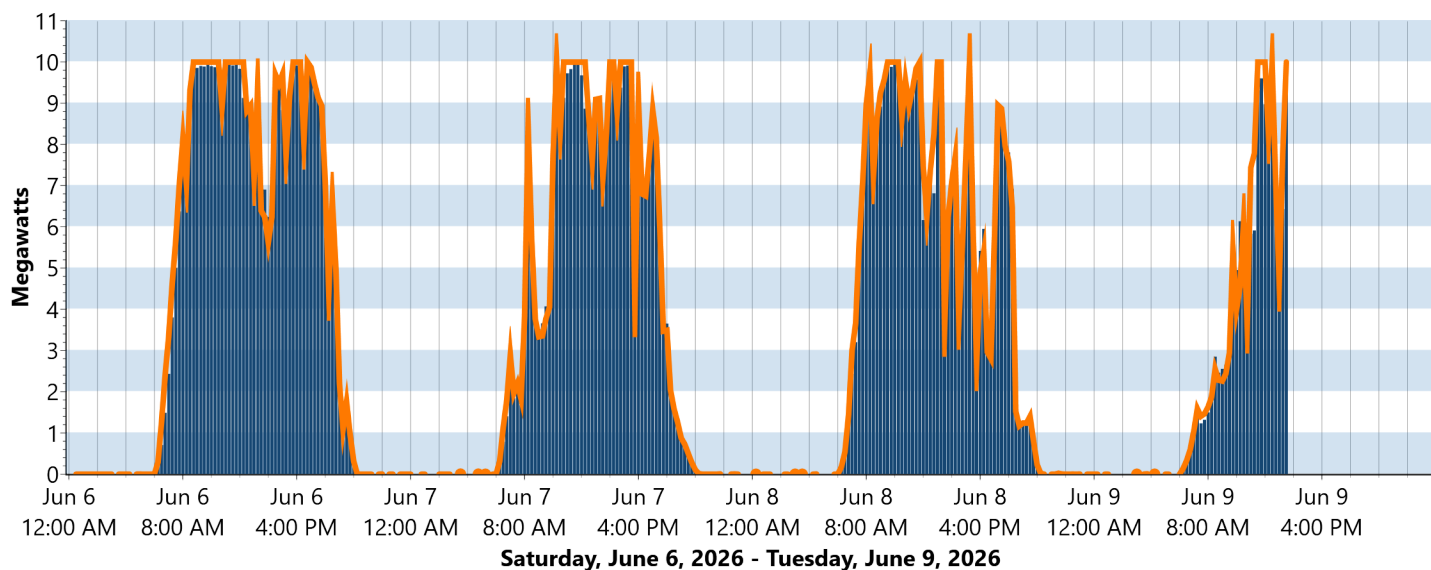


RULE TOOL RESULTS

Category	Rule	Result		Success %
Communication	Network Communication	Pass	✓	100
	Upload	Pass	✓	100
	Availability	Pass	✓	100
Configuration	System Size	Pass	✓	100
	PV Model	Pass	✓	100
	WS & PV Model	Pass	✓	100
	Default Alerts	Fail	⚠	0
	Hardware Configuration	Pass	✓	100
	Site Configuration	Pass	✓	100
	Alert Recipients	Pass	✓	100
Data	Meter vs. Inverter Energy	Pass	✓	98
	Power vs. Energy	Pass	✓	100
	Meter Voltages	Pass	✓	100
	Weather Station Data	Fail	⚠	0

PERFORMANCE

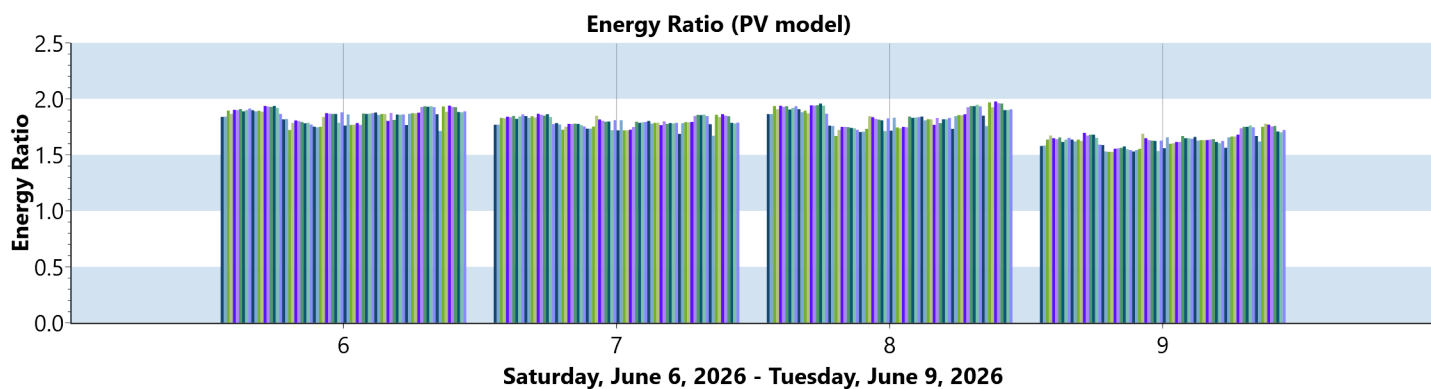
Metered Production and Model



Total Production	Expected Production*	Energy Ratio
304790 kWh	310457 kWh	98 %

*Expected production is based on Stem's PV model and weather data. PV model configuration can be checked in the Inverter hardware configuration pages.

Inverter Energy Ratio*



- INVERTER 1 ● INVERTER 2 ● INVERTER 3 ● INVERTER 4 ● INVERTER 5 ● INVERTER 6 ● INVERTER 7 ● INVERTER 8
- INVERTER 9 ● INVERTER 10 ● INVERTER 11 ● INVERTER 12 ● INVERTER 13 ● INVERTER 14 ● INVERTER 15 ● INVERTER 16
- INVERTER 17 ● INVERTER 18 ● INVERTER 19 ● INVERTER 20 ● INVERTER 21 ● INVERTER 22 ● INVERTER 23 ● INVERTER 24
- INVERTER 25 ● INVERTER 26 ● INVERTER 27 ● INVERTER 28 ● INVERTER 29 ● INVERTER 30 ● INVERTER 31 ● INVERTER 32
- INVERTER 33 ● INVERTER 34 ● INVERTER 35 ● INVERTER 36 ● INVERTER 37 ● INVERTER 38 ● INVERTER 39 ● INVERTER 40
- INVERTER 41 ● INVERTER 42 ● INVERTER 43 ● INVERTER 44 ● INVERTER 45 ● INVERTER 46 ● INVERTER 47 ● INVERTER 48
- INVERTER 49 ● INVERTER 50 ● INVERTER 51 ● INVERTER 52 ● INVERTER 53 ● INVERTER 54 ● INVERTER 55 ● INVERTER 56
- INVERTER 57 ● INVERTER 58 ● INVERTER 59 ● INVERTER 60 ● INVERTER 61 ● INVERTER 62 ● INVERTER 63 ● INVERTER 64
- INVERTER 65 ● INVERTER 66 ● INVERTER 67 ● INVERTER 68 ● INVERTER 69 ● INVERTER 70 ● INVERTER 71 ● INVERTER 72
- INVERTER 73 ● INVERTER 74 ● INVERTER 75 ● INVERTER 76 ● INVERTER 77 ● INVERTER 78 ● INVERTER 79 ● INVERTER 80

*Inverter Energy Ratio is based on Stem's PV model and weather data. PV model configuration can be checked in the Inverter hardware configuration pages.

RULE TOOL INFORMATION

Network Communication: Identifies local network failures by identifying common communication problems on a devices connected via TCP and RS-485.

Upload: Verifies that all devices have uploaded recently. Most devices are expected to upload at least once every 15 minutes. Some devices may be allowed as long as 30 hours, depending on how often they normally upload.

Availability: Verifies that there are no gaps in the 15 minute data from all devices. For devices that report data continuously, there are normally 96 data points per day. For devices that report data only when there's sunlight, the expected number of data points is calculated based on the number of hours that the sun's elevation is higher than 15 degrees.

System Size: Verifies that system size is set on the site and inverters.
For sites with inverters, verify that each inverter system size is greater than zero.
For sites without inverters, verify that the site system size is set.

PV Model: Verifies that the PV model is completely configured for every inverter.

WS & PV Model: Makes sure that irradiance and module temperature data are available from the weather stations on the site.

Default Alerts: Verify that the devices are configured with the default alerts.

Hardware Configuration: Check a number of hardware configuration details, including alerts settings, PV and DC combiner output settings and the PV models.

Site Configuration: Check a number of site configuration details, including:
Latitude, longitude, time zone, Allow alert email and Limited communication settings

Alert Recipients: Make sure alerts are setup to be generated from this site and that there is at least one alert notification recipient.

Meter vs. Inverter Energy: Verifies that the energy measurements correlate between each production meter and the inverters that are connected to it. The result is the lowest ratio of relative error between each meter and the sum of the inverters that are connected to the meter.

Power vs. Energy: Verify that the integral of the power matches the energy to within 10%. This test is run on individually on each inverter and meter.

Meter Voltages: Makes sure that the phase voltages are within 5% of each other.

Weather Station Data: Verifies that temperature and irradiance data is within expected ranges.
Checks:
Temperature data varies throughout the day.
Ambient and modules temperatures vary above and below the average locally reported ambient temperature (national weather service data).
Module temperature goes higher than ambient.
Irradiance is within 70% of expected GHI (based on virtual weather data)
Irradiance from all sensors (normalized to GHI) is within 15% of the average

Rule Tool checks the past 24 hours for pass/failure

View the complete rule tool checklist and other site information via Powertrack: apps.alsoenergy.com